

The Muscle Builder

Complete Product Workflow

Screen-to-screen activity guide

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Designed & built by Akhansha Sen

This PDF maps every primary screen and nested flow: onboarding, home, daily check-in, workout plan, live session, nutrition, progress (history + graphs), profile, and the weekly gym / rest / cheat routine.

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1. App overview and navigation map

The Muscle Builder is a local-first Flutter companion: mood-aware workouts, Indian meal suggestions, weekly schedule (gym / rest / cheat), and progress tracking with charts.

Bottom navigation (App Shell)

- Home - greeting, today routine card, plan CTA, quick links
- Workout - jump into today's plan or start check-in
- Nutrition - pre/post meals, ingredient chips, Food Pharmer pointers
- Progress - stats, week strip, analysis charts, histories
- Profile - identity, guides, weekly routine editor, settings

Full-screen nested routes (no bottom bar)

- /onboarding - first-run setup (5 steps)
- /checkin - daily check-in (5 steps) then generate plan
- /plan/:planId - customize routine and start session
- /session/:sessionId - live train / rest / finish loop

Happy-path navigation

Onboarding > Home > Check-in > Plan > Session > Progress / Nutrition

2. End-to-end daily training workflow

A. First launch

- If onboarding incomplete, app redirects to /onboarding
- User enters name, body stats, goals, equipment, diet
- On finish, profile is saved and user lands on Home

B. Typical gym day

- Home: see planned day type; optionally log Gym / Rest / Cheat
- Start or re-do Daily check-in (5 steps including pre-meal)
- Generate saves WorkoutPlan (exercises + pre/post meals)
- Open workout on Plan screen: trim moves, sets, meals, gym time
- Start session: Form cues -> log sets -> Rest -> next exercise
- Finish: session completed, day auto-marked Gym, calorie dialog
- Optional: See meals opens Nutrition on Post-workout tab
- Progress refreshes sets, sessions, day history, and graphs

C. Rest or cheat day

- On Home Today card, log Rest or Cheat
- Nutrition and Profile remain available
- Progress stores planned vs logged for adherence

3. Screen-by-screen activity

3.1 Onboarding (/onboarding)

Shown only when onboardingComplete is false.

- Step 1 - Name and journey name
- Step 2 - Body stats (weight, height, age, gender, activity)
- Step 3 - Goals, experience, coach tone
- Step 4 - Equipment multi-select
- Step 5 - Diet type, cuisine region, allergies, macros toggle
- Final CTA Start training saves profile and goes to Home

3.2 Home (/home)

- Personalized tip from coach tone
- This-week workout count and coach chip
- Today routine card: planned kind + log Gym / Rest / Cheat
- No plan: Start check-in
- Has plan: Open workout, Re-do check-in, encouragement
- Quick actions: Meals, Progress
- Pull-to-refresh for plan and week stats

Home > Check-in or Plan or Nutrition or Progress

3.3 Workout tab (/workout)

- Shortcut into the same plan / check-in loop as Home
- Opens /plan/:id when a plan exists, else /checkin

3.4 Daily check-in (/checkin) - 5 steps

- 1 Mood - tired / stressed / low / energetic / motivated
- 2 Intensity + gym minutes (20-150) driving exercise count
- 3 Focus - strength / stamina / hypertrophy / mobility
- 4 Target muscles - body map, presets, Surprise me
- 5 Pre-workout meal - diet / allergy / chip filtered
- Generate creates plan, invalidates providers, continues to training

3.5 Workout plan (/plan/:planId)

- Ordered exercise list (compounds before isolation)
- Change gym minutes to resize today's active list
- Adjust sets; skip an exercise for today
- View / swap pre and post meals; form posture gallery
- Calorie snapshot: estimated burn vs meal fuel
- Start session creates WorkoutSession and opens Session

3.6 Live session (/session/:sessionId)

- Form panel - step-by-step cues (timer paused)
- Train panel - weight, reps, RPE; tick sets (live Progress)
- Rest panel - countdown with skip option
- Finish completes session, marks today Gym, shows calorie balance
- See meals -> /nutrition?timing=post

3.7 Nutrition (/nutrition)

- Tabs: Pre-workout / Post-workout
- Cuisine chips: North / South / Pan Indian
- Macros on/off persisted on profile
- I want to eat chips (horizontal): protein, curd, eggs, paneer, chicken, oats, dates, fruit, spinach, ragi/millet, legumes
- Protein focus shortcut
- Food Pharmer pointer cards open foodpharmer.health
- Meal cards: portion, timing, macros, nutrition notes, recipes

3.8 Progress (/progress)

- Encouragement + today's calorie balance
- Stats: sets today/week, workouts done/this week
- Muscles trained from completed sessions
- This week's plan strip (7 day kinds)
- Routine analysis: adherence %, pie chart, gym-days bar chart
- Day history (planned vs logged)
- Pre-meal log from check-ins
- Session history (muscles, duration, sets, pre meal)

3.9 Profile (/profile)

- Journey hero with editable names / stats
- Yogurt and dahi fuel carousel -> Nutrition
- Coach-approved reading (Food Pharmer, ACSM, EatRight, NHS, NIN, IIMR)
- Weekly routine editor (Mon-Sun Gym / Rest / Cheat)
- Theme, appearance, coach personality
- Training and diet settings
- Faded signature: Designed & built by Akhansha Sen

4. Weekly routine (gym / rest / cheat)

This calendar is separate from the exercise generator (RoutineEngine).

Define - Profile

- Profile > Settings > Weekly routine
- Assign each weekday Gym, Rest, or Cheat
- Stored in WeeklyRoutines (JSON weekday map)
- Default: Mon/Wed/Fri/Sat Gym, Tue/Thu Rest, Sun Cheat

Execute - Home

- Today card shows Planned kind from the template
- User logs actual kind into DayLogs
- Finishing a session auto-logs Gym for today

Analyze - Progress

- Week strip uses logged actual, else planned
- Adherence = logged gym hits / planned gym days
- Pie: gym / rest / cheat mix (about 4 weeks)
- Bar: logged gym days per week across 4 weeks
- Day history lists planned vs logged

5. Nutrition and Food Pharmer

- Meals from assets/data/meals.json including ragi, spinach, dates, fruits
- Filters: timing x diet x allergies x region x ingredient tags (OR)
- nutritionNotes cite ICMR-NIN / IIMR / USDA-style facts (educational)
- Food Pharmer cards deep-link to Better for You criteria pages
- Unlock recipe stores meal id on profile

6. Progress, history, and graphs

- Each logged set bumps a tick provider so Progress refreshes live
- Completed sessions + set logs drive workout stats
- Pre/post meals on plans feed the Pre-meal log
- DayLogs + WeeklyRoutines feed strip, history, and fl_chart charts

7. Data that persists between screens

Drift / SQLite tables

- UserProfiles - identity, diet, theme, chips, unlocked recipes
- DailyCheckIns - mood, intensity, muscles, focus
- WorkoutPlans - exercises, encouragement, meals, gym minutes
- WorkoutSessions / SetLogs - training history
- WeeklyRoutines - Mon-Sun template
- DayLogs - per-date planned + actual day kind

Cross-screen effects

- Profile diet / allergies / chips -> Nutrition + check-in meals
- Check-in -> Plan -> Session -> Progress + day=Gym
- Weekly routine -> Home today + Progress graphs
- Session finish -> optional Nutrition post tab

8. Route quick reference

Route	Screen	Notes
/onboarding	OnboardingScreen	First run
/home	HomeScreen	Tab 1
/workout	WorkoutTabScreen	Tab 2
/nutrition	NutritionScreen	Tab 3 (+ timing query)
/progress	ProgressScreen	Tab 4
/profile	ProfileScreen	Tab 5
/checkin	CheckInScreen	Nested full screen
/plan/:planId	WorkoutPlanScreen	Nested full screen
/session/:sessionId	SessionScreen	Nested full screen

Master activity flowchart

Onboarding > Home (log today) > Check-in (5 steps) > Plan (customize) > Session (form / train / rest) > Finish (Gym day + calories) > Progress (history / graphs) and/or Nutrition (post meals). Use Profile anytime for weekly routine and diet preferences.

Not medical advice. External nutrition links are educational attributions only.